

Engineering Physics Lab

Engineering Physics Lab is a laboratory that focuses on the development of skills in different areas of Physics. The experiments are designed to illustrate phenomena and expose students to measuring instruments . The long-term goal of the current work in the lab is to develop a spacecraft that could be self-sufficient and support research experiments on the moon, Mars, and beyond . Another team is working to develop a robotic arm for the spacecraft that could collect samples from the surface of a planet or asteroid .

The Project Lab is the core of innovation in Engineering Physics, and the main resource for our students to learn the technical and leadership aspects of becoming professional engineers . With their unique combination of engineering breadth and deep analytic training, our students are exceptionally well .

An Engineering Physics degree prepares students to work in the private sector or in national laboratories at the very forefront of technology, or to pursue an advanced degree in engineering . An Engineering Physics degree also prepares students to pursue an advanced degree in physics; other engineering majors do not .

Engineering Physics labs include experiments from Physics I and II . Physics Explorations and Projects is a collection of laboratory investigations aligned to the Next Generation Science Standards (NGSS) . Most included investigations set up a situation for students to explore and analyze with some guidance from the instructor—a guided-inquiry format that is very different from traditional laboratory .

List of Experiments

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observing aggressive behavior, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.

9. **The Invisible Gorilla:** A study on inattention blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Trolley Problem:** A thought experiment in ethics, first introduced by Philippa Foot in 1967.

Any ten experiments (at least four from each group)

Group 1: Physics

1. **The Double-Slit Experiment:** This experiment demonstrates the wave-particle duality of light, where light behaves as both a wave and a particle.
2. **The Michelson-Morley Experiment:** This experiment was designed to detect the presence of the hypothetical luminiferous aether, a medium through which light was thought to propagate.
3. **The Millikan Oil Drop Experiment:** This experiment measured the charge of an electron by balancing the gravitational and electrical forces on tiny charged droplets of oil suspended between two metal electrodes.
4. **The Rutherford Gold Foil Experiment:** This experiment demonstrated the existence of a small, dense, positively charged nucleus at the center of an atom.

Group 2: Chemistry

1. **The Iodine Clock Reaction:** This experiment demonstrates the concept of reaction rates and how they can be affected by factors such as temperature and concentration.
2. **The Briggs-Rauscher Oscillating Reaction:** This experiment demonstrates the concept of chemical oscillations, where the concentrations of reactants and products oscillate over time.
3. **The Belousov-Zhabotinsky Reaction:** This experiment demonstrates the concept of chemical oscillations and the emergence of complex patterns in a chemical system.
4. **The Elephant Toothpaste Experiment:** This experiment demonstrates the decomposition of hydrogen peroxide into water and oxygen gas, catalyzed by potassium iodide.

Group 3: Biology

1. **The Miller-Urey Experiment:** This experiment simulated the conditions thought to be present on the early Earth and tested the hypothesis that the building blocks of life could have formed spontaneously under these conditions.
2. **The Hershey-Chase Experiment:** This experiment demonstrated that DNA, not protein, is the genetic material of life.

Group A

Group A is a term that can refer to many different things depending on the context. Here are some possible meanings of Group A:

1. In the context of a study or experiment, Group A may refer to the first group of participants or subjects.
2. In sports, Group A may refer to the first group or division of teams in a tournament or league.
3. In the context of a classification system, Group A may refer to the first group or category of items or individuals.
4. In the context of a project or organization, Group A may refer to the first group or team of individuals working on a specific task or objective.

1. To determine the wavelength of sodium light by Newton's ring experiment.

- Newton's ring experiment is a method to determine the wavelength of light using the phenomenon of interference.
- In this experiment, a plano-convex lens is placed on a flat glass plate, and a monochromatic light source, such as sodium light, is used to illuminate the setup.
- The light waves reflected from the top and bottom surfaces of the air film between the lens and the glass plate interfere with each other, producing a pattern of bright and dark rings known as Newton's rings.
- The radius of the rings can be measured using a traveling microscope, and the wavelength of the light can be calculated using the formula: $\lambda = \frac{D_n^2 - D_{n-m}^2}{4R}$, where λ is the wavelength of the light, D_n is the diameter of the n th dark ring, R is the radius of curvature of the lens, and m is the number of rings between the two measured rings.
- By measuring the diameters of multiple rings and using the above formula, the wavelength of the sodium light can be accurately determined.

2. To determine the wavelength of different spectral lines of mercury light using plane

- Spectral lines are produced when an element is heated and its electrons are excited to higher energy levels.
- When the electrons return to their ground state, they emit light at specific wavelengths, producing a characteristic spectrum for each element.
- Mercury is a commonly used element in spectroscopy due to its well-defined and easily observable spectral lines.
- To determine the wavelength of different spectral lines of mercury light, a plane transmission diffraction grating can be used.
- A diffraction grating is an optical component with a regular pattern of closely spaced lines that diffract light into its component wavelengths.
- When light from a mercury lamp is passed through a diffraction grating, the light is diffracted into its component wavelengths, producing a spectrum.

- By measuring the angles at which the different spectral lines are observed, the wavelengths of the spectral lines can be calculated using the grating equation.
- The grating equation relates the angle of diffraction, the grating spacing, and the wavelength of the diffracted light.
- By using this equation and the measured angles, the wavelengths of the different spectral lines of mercury light can be determined.

Transmission Grating

- Transmission Gratings are designed to disperse incident light at specific angles. Angular dispersion is a function of angle of incidence and groove spacing. Dispersion increases as angle of incidence increases or as groove spacing decreases.
- Transmission gratings are manufactured in the same way as ruled reflection gratings in which a replica of a master grating is created by duplicating its grooved surface with a thin, vacuum deposition-coated layer, except they have no reflective overcoat.
- Transmission gratings offer low polarization sensitivity as opposed to reflection gratings because incident light is not reflected by a mirror coating such as aluminum. Since light is transmitted through the grating, transmission gratings can be used in compact, in-line configurations.
- Transmission gratings are diffraction gratings which are used in transmission. This is in contrast to the more common reflection gratings, which reflect all incident light, as far as it is not absorbed or scattered. Transmission gratings can be volume Bragg gratings or surface relief gratings.
- The product line of transmission gratings by Coherent utilizes state-of-the-art design and precise lithographic wafer-scale fabrication to deliver record-breaking optical performance.

3. To determine the specific rotation of cane sugar solution using polarimeter.

A polarimeter is an instrument used to measure the angle of rotation of plane-polarized light as it passes through an optically active substance. The specific rotation of a substance is the angle of rotation of plane-polarized light per unit path length and per unit concentration of the substance.

To determine the specific rotation of a cane sugar solution using a polarimeter, the following steps can be followed:

1. Prepare a solution of cane sugar by dissolving a known mass of cane sugar in a known volume of water.
2. Fill a polarimeter tube with the cane sugar solution and place it in the polarimeter.
3. Adjust the polarimeter to obtain a reading of the angle of rotation of the plane-polarized light as it passes through the cane sugar solution.

4. Calculate the specific rotation of the cane sugar solution using the formula:
$$\text{specific rotation} = (\text{observed rotation}) / (\text{concentration} \times \text{path length}).$$

The specific rotation of the cane sugar solution can be used to determine the concentration of the solution, or to identify the presence of other optically active substances in the solution. It is important to note that the specific rotation of a substance is dependent on the wavelength of the light used, the temperature of the solution, and the solvent used to prepare the solution. These factors should be controlled and reported when determining the specific rotation of a substance.

4. To determine the focal length of the combination of two lenses separated by a distance

The focal length of the combination of two lenses separated by a distance can be determined using the formula:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

Where: - f is the focal length of the combination of two lenses - f_1 is the focal length of the first lens - f_2 is the focal length of the second lens - d is the distance between the two lenses

To determine the focal length of the combination of two lenses, the following steps can be followed: 1. Measure the focal length of the first lens (f_1) using a suitable method. 2. Measure the focal length of the second lens (f_2) using a suitable method. 3. Measure the distance between the two lenses (d). 4. Substitute the values of f_1 , f_2 , and d in the formula to calculate the focal length of the combination of two lenses (f).

It is important to note that the formula is only valid when the distance between the two lenses is much smaller than the focal lengths of the lenses. If the distance between the lenses is large, the formula may not give accurate results. In such cases, other methods may be used to determine the focal length of the combination of two lenses.

Verification of the formula for the focal length of combination of lenses

The formula for the focal length of a combination of lenses is given by:

$$1/f = 1/f_1 + 1/f_2 - d/f_1f_2$$

where f is the focal length of the combination of lenses, f_1 and f_2 are the focal lengths of the individual lenses, and d is the distance between the two lenses.

To verify this formula, we can perform an experiment using a convex lens and a concave lens. The steps for the experiment are as follows:

1. Place a convex lens of known focal length f_1 on an optical bench.

2. Place a concave lens of known focal length f_2 at a distance d from the convex lens.
3. Place an object at a distance greater than f_1 from the convex lens.
4. Adjust the position of the screen until a sharp image of the object is formed on it.
5. Measure the distance between the object and the screen. This distance is the image distance v .
6. Measure the distance between the object and the convex lens. This distance is the object distance u .
7. Using the lens formula $1/f = 1/u + 1/v$, calculate the focal length f of the combination of lenses.
8. Compare the calculated value of f with the value obtained using the formula for the focal length of a combination of lenses.

The experimentally obtained value of f should be in close agreement with the value obtained using the formula for the focal length of a combination of lenses, thus verifying the formula.

5. To measure attenuation in an optical fiber

Attenuation is the reduction in the strength of an optical signal as it travels through an optical fiber. It is measured in decibels per kilometer (dB/km). The following are the steps to measure attenuation in an optical fiber:

1. **Set up the equipment:** Set up an optical power meter and a light source at opposite ends of the fiber. The light source should be set to the same wavelength as the system that will be used with the fiber.
2. **Measure the reference power:** Turn on the light source and measure the power level at the power meter. This is the reference power level.
3. **Disconnect the fiber:** Disconnect the fiber from the light source and connect it to the power meter.
4. **Measure the output power:** Turn on the light source and measure the power level at the power meter. This is the output power level.
5. **Calculate the attenuation:** The attenuation is calculated by subtracting the output power from the reference power and dividing by the length of the fiber in kilometers. The result is expressed in dB/km.

Attenuation is an important parameter to consider when designing an optical fiber communication system. It determines the maximum distance that the signal can travel before it needs to be amplified or regenerated. A low attenuation fiber is desirable for long-distance communication.

6. To determine the wavelength of He-Ne laser light using single slit diffraction.

1. The experiment is set up by shining a He-Ne laser light through a single slit and onto a screen.
2. The diffraction pattern produced on the screen is observed and measured.
3. The distance between the central maximum and the first minimum on either side is measured.
4. The slit width and the distance between the slit and the screen are also measured.
5. Using the formula for single slit diffraction, the wavelength of the He-Ne laser light can be calculated.
6. The formula for single slit diffraction is: $\lambda = (a * \sin \theta) / m$, where λ is the wavelength, a is the slit width, θ is the angle of diffraction, and m is the order of the minimum.
7. By substituting the measured values into the formula, the wavelength of the He-Ne laser light can be determined.

7. To study the polarization of light using He-Ne laser light.

Polarization is a property of light that describes the direction of its electric field. Light can be polarized in several ways, including linear, circular, and elliptical polarization.

A He-Ne laser is a type of gas laser that uses a mixture of helium and neon gases as the lasing medium. The light emitted by a He-Ne laser is typically linearly polarized.

To study the polarization of light using a He-Ne laser, the following steps can be followed:

1. Set up the He-Ne laser and turn it on.
2. Place a polarizer in the path of the laser beam.
3. Rotate the polarizer and observe the changes in the intensity of the laser light.
4. Record the intensity of the laser light as a function of the angle of the polarizer.
5. Analyze the data to determine the polarization state of the laser light.

By following these steps, one can study the polarization of light using a He-Ne laser. This can provide valuable insights into the properties of light and its interaction with matter.

8. To determine the wavelength of sodium light with the help of Fresnel's bi-prism.

Fresnel's bi-prism is an optical instrument used to determine the wavelength of light. It consists of two prisms placed base to base, forming a small angle between them. When a beam of light passes through the bi-prism, it is split into two beams, which interfere with each other to produce a pattern of bright and dark fringes.

To determine the wavelength of sodium light using Fresnel's bi-prism, the following steps can be followed:

1. Set up the bi-prism experiment by placing the bi-prism in front of a sodium light source and a screen at a suitable distance.
2. Adjust the position of the bi-prism and the screen until a clear interference pattern is observed on the screen.
3. Measure the distance between the screen and the bi-prism, as well as the distance between the central bright fringe and a fringe of known order.
4. Use the formula for the fringe width to calculate the wavelength of the sodium light.

The formula for the fringe width is given by: $w = \frac{\lambda D}{d}$, where w is the fringe width, λ is the wavelength of the light, D is the distance between the screen and the bi-prism, and d is the distance between the two virtual sources formed by the bi-prism.

By following these steps and using the formula, the wavelength of sodium light can be accurately determined using Fresnel's bi-prism.

9. To determine the coefficient of viscosity of a given liquid.

Viscosity is a measure of a fluid's resistance to flow. The coefficient of viscosity is a measure of the internal friction of a fluid. There are several methods to determine the coefficient of viscosity of a given liquid, including:

1. **Ostwald viscometer:** This method involves measuring the time it takes for a fixed volume of liquid to flow through a narrow tube under the influence of gravity. The coefficient of viscosity can be calculated using Poiseuille's equation.
2. **Falling sphere viscometer:** This method involves measuring the terminal velocity of a sphere falling through a liquid. The coefficient of viscosity can be calculated using Stokes' law.
3. **Rotational viscometer:** This method involves measuring the torque required to rotate a spindle at a constant speed in a liquid. The coefficient of viscosity can be calculated using the relationship between torque, spindle speed, and the dimensions of the spindle and liquid container.

Each of these methods has its own advantages and limitations, and the choice of method may depend on factors such as the type of liquid being tested and the desired level of accuracy. It is important to carefully follow the experimental procedure and to take multiple measurements to ensure accurate results.

10. To determine the value of acceleration due to gravity (g) using compound pendulum.

The compound pendulum is an experimental setup used to determine the value of acceleration due to gravity (g). The setup consists of a rigid body suspended

from a fixed horizontal axis, which is allowed to oscillate freely under the influence of gravity. The time period of oscillation of the compound pendulum is given by the formula $T = 2\sqrt{I/mgd}$, where I is the moment of inertia of the body about the axis of suspension, m is the mass of the body, g is the acceleration due to gravity, and d is the distance between the center of mass of the body and the axis of suspension.

To determine the value of g using the compound pendulum, the following steps are followed:

1. The compound pendulum is set up and allowed to oscillate freely.
2. The time period of oscillation is measured using a stopwatch.
3. The mass of the body and the distance between the center of mass and the axis of suspension are measured.
4. The moment of inertia of the body about the axis of suspension is calculated using the parallel axis theorem.
5. The value of g is calculated using the formula $g = 4\pi^2(I/mdT^2)$.

This experiment can be repeated with different masses and distances to obtain more accurate results. The value of g obtained from this experiment is an approximation, as it assumes that the oscillations of the compound pendulum are small and that air resistance is negligible. However, it provides a simple and effective way to determine the value of acceleration due to gravity.

Group B

Group B is a term that can refer to many different things depending on the context. Here are some possible interpretations of the term “Group B”:

1. In the context of rally racing, Group B was a set of regulations introduced in 1982 for competition vehicles in sportscar racing and rallying regulated by the Fédération Internationale de l'Automobile (FIA).
2. In the context of microbiology, Group B Streptococcus (GBS) is a type of bacterial infection that can be found in a pregnant woman's vagina or rectum and can be passed to the baby during delivery.
3. In the context of vitamins, Group B refers to the B vitamins, a group of water-soluble vitamins that play important roles in cell metabolism.

1. To determine the energy band gap of a given semiconductor material:

- The energy band gap of a semiconductor material is the difference in energy between the top of the valence band and the bottom of the conduction band.
- The band gap is a crucial property of semiconductors as it determines their electrical conductivity and optical properties.
- There are several methods to determine the energy band gap of a semiconductor material, including optical absorption, photoluminescence, and

electrical measurements.

- In the optical absorption method, the absorption coefficient of the material is measured as a function of the photon energy. The absorption edge, where the absorption coefficient increases rapidly, corresponds to the band gap energy.
- In the photoluminescence method, the material is excited with light of a known energy, and the emitted light is analyzed to determine the band gap energy.
- In electrical measurements, the current-voltage characteristics of the material are measured at different temperatures. The temperature dependence of the current can be used to determine the band gap energy.
- The choice of method depends on the specific material and the desired accuracy of the measurement.

2. To study Hall effect and determine Hall coefficient, carrier density and mobility of a

The Hall effect is a phenomenon that occurs when a magnetic field is applied perpendicular to the flow of electric current in a conductor. This causes a voltage difference to develop across the conductor, perpendicular to both the current and the magnetic field. This voltage difference is known as the Hall voltage.

The Hall coefficient, carrier density, and mobility of a material can be determined by measuring the Hall voltage and the current flowing through the material. The Hall coefficient is a measure of the strength of the Hall effect in a material and is related to the carrier density and mobility of the material.

The carrier density refers to the number of charge carriers (such as electrons or holes) in a material per unit volume. The mobility of a material is a measure of how easily the charge carriers can move through the material when an electric field is applied.

To determine the Hall coefficient, carrier density, and mobility of a material, the following steps can be followed:

1. Prepare a sample of the material in the form of a thin rectangular strip.
2. Connect the sample to a circuit that allows a current to flow through it.
3. Apply a magnetic field perpendicular to the flow of current in the sample.
4. Measure the Hall voltage that develops across the sample.
5. Use the measured values of the current, magnetic field, and Hall voltage to calculate the Hall coefficient, carrier density, and mobility of the material using the appropriate equations.

By studying the Hall effect and determining the Hall coefficient, carrier density, and mobility of a material, it is possible to gain a better understanding of the material's electrical properties and behavior. This information can be useful in the design and development of electronic devices and systems.

Given semiconductor material using Hall effect setup

The Hall effect is a phenomenon that occurs in a conductor or semiconductor when a magnetic field is applied perpendicular to the flow of current. This causes a voltage difference to develop across the material, known as the Hall voltage. The Hall effect can be used to determine several properties of a given semiconductor material, including:

1. **Carrier concentration:** The Hall voltage is directly proportional to the carrier concentration in the material. By measuring the Hall voltage, the carrier concentration can be determined.
2. **Type of semiconductor:** The sign of the Hall voltage can be used to determine whether the semiconductor is a p-type or n-type material. A positive Hall voltage indicates a p-type material, while a negative Hall voltage indicates an n-type material.
3. **Mobility:** The Hall voltage is also related to the mobility of the charge carriers in the material. By measuring the Hall voltage and the carrier concentration, the mobility can be calculated.

To perform a Hall effect measurement on a given semiconductor material, a Hall effect setup is used. This typically consists of a sample of the material, a magnetic field source, and a voltmeter to measure the Hall voltage. The sample is placed in the magnetic field, and a current is passed through it. The Hall voltage is then measured across the sample, and the above properties can be determined.

3. To determine the variation of magnetic field with the distance along the axis of a current

- The magnetic field produced by a current-carrying wire can be determined using the Biot-Savart law.
- The Biot-Savart law states that the magnetic field at a point due to a small current element is directly proportional to the current, the length of the current element, and inversely proportional to the square of the distance between the point and the current element.
- The magnetic field along the axis of a current-carrying wire can be determined by integrating the magnetic field due to small current elements along the length of the wire.
- The magnetic field decreases as the distance from the wire increases.
- The direction of the magnetic field can be determined using the right-hand rule.
- The strength of the magnetic field can be measured using a Hall probe or a magnetometer.
- The variation of the magnetic field with distance can be plotted on a graph to show the relationship between the two.

Carrying Coil and Estimate the Radius of the Coil

A carrying coil is a coil of wire that carries an electric current. The magnetic field produced by the current in the coil can be used to estimate the radius of the coil. Here are the steps to estimate the radius of a carrying coil:

1. Measure the magnetic field strength at the center of the coil using a Gauss-meter.
2. Determine the number of turns in the coil.
3. Use the formula $B = (\mu_0 * I * N) / (2 * R)$ to calculate the radius of the coil, where B is the magnetic field strength, μ_0 is the permeability of free space, I is the current in the coil, N is the number of turns in the coil, and R is the radius of the coil.
4. Solve the equation for R to find the estimated radius of the coil.

4. To verify Stefan's law by electric method

Stefan's law states that the total energy radiated per unit surface area of a black body across all wavelengths per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature.

To verify Stefan's law by electric method, the following steps can be taken:

1. Set up the apparatus, which includes a Leslie cube, a thermopile, a galvanometer, and a low voltage bulb.
2. Fill the Leslie cube with boiling water and place it on a heatproof mat.
3. Place the thermopile in front of one of the faces of the Leslie cube, at a fixed distance.
4. Connect the thermopile to the galvanometer.
5. Switch on the low voltage bulb and adjust its position so that it illuminates the face of the Leslie cube.
6. Record the deflection of the galvanometer.
7. Repeat the experiment with the thermopile at different distances from the Leslie cube.
8. Plot a graph of the galvanometer deflection against the inverse square of the distance between the thermopile and the Leslie cube.
9. The graph should be a straight line, verifying Stefan's law.

5. To determine resistance per unit length and specific resistance of a given resistance

- Resistance is a measure of the opposition to the flow of electric current through a conductor.
- The resistance of a conductor depends on its length, cross-sectional area, and the material it is made of.
- The resistance per unit length is calculated by dividing the total resistance of the conductor by its length.

- The specific resistance, also known as resistivity, is a property of the material the conductor is made of. It is defined as the resistance of a conductor of unit length and unit cross-sectional area.
- To determine the resistance per unit length and specific resistance of a given resistance, the following steps can be followed:
 1. Measure the length of the conductor using a ruler or measuring tape.
 2. Measure the cross-sectional area of the conductor using a micrometer screw gauge or Vernier calipers.
 3. Connect the conductor in a circuit with a battery, an ammeter, and a voltmeter.
 4. Measure the current flowing through the conductor using the ammeter and the potential difference across the conductor using the voltmeter.
 5. Calculate the resistance of the conductor using Ohm's law, which states that the resistance is equal to the potential difference divided by the current.
 6. Divide the resistance of the conductor by its length to obtain the resistance per unit length.
 7. Divide the resistance of the conductor by its cross-sectional area to obtain the specific resistance or resistivity of the material it is made of.

Using Carey Foster's Bridge

Carey Foster's Bridge is an electrical circuit used to measure medium resistances or small differences between two large resistances. It is a variant of the Wheatstone bridge, invented by Carey Foster .

- **Construction:** The construction of Carey Foster's bridge circuit is similar to the Wheatstone bridge, which consists of four arms in which four resistances are connected .
- **Working:** Carey Foster bridge measures fundamentally the difference between the resistance of two elements .
- **Applications:** Carey Foster bridge can calculate medium resistances or can compare and measure the two large/equal resistance values with small variations .

6. To study the resonance condition of a series LCR circuit.

A series LCR circuit consists of an inductor (L), a capacitor (C), and a resistor (R) connected in series. The resonance condition of a series LCR circuit occurs when the inductive reactance (XL) is equal to the capacitive reactance (XC). At resonance, the circuit behaves as a purely resistive circuit, and the current in the circuit is maximum.

The resonance condition can be mathematically expressed as:

$$XL = XC$$

where $X_L = 2 \pi f L$ and $X_C = 1 / (2 \pi f C)$

Substituting the values of X_L and X_C in the above equation, we get:

$$2 \pi f L = 1 / (2 \pi f C)$$

Solving for f , we get the resonant frequency (f_r) as:

$$f_r = 1 / (2 \pi \sqrt{LC})$$

At resonance, the impedance (Z) of the circuit is equal to the resistance (R) of the circuit, and the current (I) in the circuit is given by:

$$I = V / Z = V / R$$

where V is the applied voltage.

The resonance condition of a series LCR circuit can be studied by varying the frequency of the applied voltage and measuring the current in the circuit. At resonance, the current in the circuit will be maximum. The resonant frequency can be calculated using the above equation and compared with the experimentally obtained value.

In summary, the resonance condition of a series LCR circuit occurs when the inductive reactance is equal to the capacitive reactance, resulting in a purely resistive circuit with maximum current. The resonant frequency can be calculated using the equation $f_r = 1 / (2 \pi \sqrt{LC})$, and the resonance condition can be studied experimentally by varying the frequency of the applied voltage and measuring the current in the circuit.

7. To determine the electrochemical equivalent (ECE) of copper.

The electrochemical equivalent (ECE) of a substance is the mass of the substance deposited or dissolved at an electrode during electrolysis by the passage of one coulomb of electric charge. The ECE of copper can be determined by performing an experiment using a copper voltameter.

1. A copper voltameter consists of two copper electrodes immersed in a solution of copper sulfate.
2. A known amount of electric charge is passed through the solution using a constant current power supply.
3. The mass of copper deposited on the cathode is measured before and after the experiment.
4. The difference in mass gives the amount of copper deposited.
5. The ECE of copper is calculated by dividing the mass of copper deposited by the amount of electric charge passed.

This experiment can be repeated with different amounts of electric charge to obtain an average value for the ECE of copper. The ECE of copper can also be calculated theoretically using Faraday's laws of electrolysis. The theoretical and experimental values can be compared to evaluate the accuracy of the experiment.

8. To calibrate the given ammeter and voltmeter by potentiometer.

Calibration is the process of comparing the readings of an instrument with a known standard to determine the accuracy of the instrument. In this experiment, we will calibrate an ammeter and a voltmeter using a potentiometer.

The following steps can be followed to calibrate the given ammeter and voltmeter by potentiometer:

1. Connect the ammeter in series with a battery and a resistor of known resistance.
2. Measure the current flowing through the circuit using the ammeter.
3. Connect the potentiometer in parallel with the ammeter and adjust the potentiometer until the potential difference across the ammeter is zero.
4. Measure the current flowing through the circuit using the potentiometer.
5. Compare the readings of the ammeter and the potentiometer to determine the accuracy of the ammeter.
6. Repeat the above steps for the voltmeter by connecting it in parallel with the battery and measuring the potential difference across the battery.
7. Adjust the potentiometer until the potential difference across the voltmeter is zero and measure the potential difference using the potentiometer.
8. Compare the readings of the voltmeter and the potentiometer to determine the accuracy of the voltmeter.

By following these steps, the given ammeter and voltmeter can be calibrated using a potentiometer. This will help to ensure that the readings of the ammeter and voltmeter are accurate and reliable.

9. To draw hysteresis (B-H curve) of a specimen in the form of a transformer and to

Hysteresis is the dependence of the state of a system on its history. In the context of magnetic materials, hysteresis refers to the lag between the magnetic field B and the magnetizing force H . The B-H curve, also known as the hysteresis loop, is a graphical representation of this phenomenon.

To draw the B-H curve of a specimen in the form of a transformer, the following steps can be followed:

1. Set up the transformer with the specimen as the core, and connect it to an AC power source.
2. Connect an ammeter in series with the primary coil to measure the current, and a voltmeter across the secondary coil to measure the induced voltage.
3. Gradually increase the current in the primary coil and record the corresponding values of the induced voltage in the secondary coil.
4. Plot the induced voltage (B) against the current (H) to obtain the B-H curve.

The shape of the B-H curve depends on the magnetic properties of the specimen. Materials with a large hysteresis loop, such as ferromagnetic materials, exhibit a large amount of energy loss when subjected to alternating magnetic fields. This energy loss is known as hysteresis loss and is an important factor to consider in the design of transformers and other magnetic devices.

Determine its hysteresis loss

Hysteresis loss is the energy loss that occurs in a magnetic material due to the reversal of magnetization. It is an important factor to consider when designing magnetic devices such as transformers and electric motors.

To determine the hysteresis loss of a magnetic material, the following steps can be followed:

1. Obtain a sample of the magnetic material and measure its dimensions.
2. Subject the sample to a cyclic magnetic field and measure the magnetic flux density at various points in the cycle.
3. Plot the magnetic flux density against the magnetic field strength to obtain the hysteresis loop.
4. Calculate the area enclosed by the hysteresis loop. This area represents the energy loss per cycle due to hysteresis.
5. Multiply the area of the hysteresis loop by the frequency of the magnetic field to obtain the hysteresis loss.

It is important to note that the hysteresis loss is dependent on the frequency of the magnetic field, the shape and size of the hysteresis loop, and the properties of the magnetic material. Different materials will have different hysteresis losses, and the loss can be minimized by selecting materials with low coercivity and high permeability.

10. To measure high resistance by leakage method.

The leakage method is a technique used to measure high resistance values. This method is based on the principle of measuring the leakage current flowing through the insulation of a conductor. The following are the steps involved in measuring high resistance by the leakage method:

1. The conductor whose resistance is to be measured is connected to a high voltage source.
2. The leakage current flowing through the insulation of the conductor is measured using a sensitive ammeter.
3. The resistance value is calculated by dividing the applied voltage by the measured leakage current.

This method is commonly used in the testing of insulation resistance of cables and other electrical equipment. It is important to note that the accuracy of the measurement depends on the sensitivity of the ammeter used and the stability of the applied voltage.

Course outcomes:

- Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course.
- These outcomes are used to assess the effectiveness of the course and to ensure that the course content aligns with the program's goals and objectives.
- Course outcomes should be specific, measurable, and achievable within the timeframe of the course.
- They should be written in clear and concise language and should be easily understood by students, instructors, and other stakeholders.
- Course outcomes should be aligned with the course content and should be used to guide the development of assessments and instructional materials.
- They should be regularly reviewed and updated to ensure that they remain relevant and accurate.
- Course outcomes can be used to evaluate student learning and to provide feedback to students on their progress.
- They can also be used to identify areas where students may need additional support or instruction.
- Course outcomes are an important tool for ensuring that courses are effective and that students are achieving the desired learning outcomes.

At the end of the course, students will be able to:

1. Demonstrate an understanding of the key concepts and principles of the subject matter.
2. Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Communicate effectively using the language and terminology of the subject matter.
4. Work collaboratively with others to achieve common goals.
5. Think critically and creatively to analyze and evaluate information and ideas.
6. Demonstrate a commitment to ongoing learning and professional development.

CO CO Statement Bloom's

CO CO Statement Bloom's is a topic that refers to the use of Bloom's Taxonomy in the creation of Course Outcome (CO) statements. Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used by educators to design assessments, curriculum, and instructional methods.

Here are some key points to consider when using Bloom's Taxonomy to create CO statements:

1. Bloom's Taxonomy consists of six levels: Remembering, Understanding,

Applying, Analyzing, Evaluating, and Creating. Each level represents a different type of cognitive process, with Remembering being the most basic and Creating being the most complex.

2. When creating CO statements, it is important to consider the level of Bloom's Taxonomy that is most appropriate for the course and its objectives. For example, a course that focuses on basic knowledge and comprehension may have CO statements that align with the Remembering and Understanding levels of Bloom's Taxonomy.
3. CO statements should be specific, measurable, and achievable. They should clearly state what the student is expected to know or be able to do by the end of the course.
4. It is important to align the CO statements with the course content, instructional methods, and assessments. This ensures that the course is designed to effectively support student learning and achievement of the stated outcomes.

In summary, CO Statement Bloom's refers to the use of Bloom's Taxonomy in the creation of Course Outcome statements. This framework can be a useful tool for educators in designing effective courses that support student learning and achievement.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness.
- Laser levels project a laser beam to create a visual reference for levelness.
- Water levels use the principle that water will always find its own level to determine levelness.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing objects such as shelves, cabinets, and picture frames to ensure they are straight and level.

CO-1 Apply the principle of interference and diffraction to find the wavelength of

1. Interference and diffraction are two phenomena that can be used to determine the wavelength of light.
2. Interference occurs when two or more waves overlap and combine to form a new wave. The resulting wave can have a larger or smaller amplitude than the individual waves, depending on the relative phase of the waves.

3. Diffraction is the bending of waves around obstacles or through openings. The amount of bending depends on the size of the opening or obstacle relative to the wavelength of the wave.
4. To find the wavelength of light using interference, one can use a device called a Michelson interferometer. This device splits a beam of light into two beams, which are then reflected back and recombined. By adjusting the path length of one of the beams, the relative phase of the two beams can be changed, causing the interference pattern to change. By measuring the change in the interference pattern, the wavelength of the light can be determined.
5. To find the wavelength of light using diffraction, one can use a device called a diffraction grating. This is a surface with many closely spaced parallel lines. When light is shone on the grating, it diffracts into many different directions. By measuring the angles of the diffracted beams, the wavelength of the light can be determined.
6. Both interference and diffraction can be used to accurately determine the wavelength of light. These principles are widely used in scientific research and have many practical applications.

Monochromatic and Polychromatic Light

- **Monochromatic light** refers to light that consists of a single wavelength or color. This type of light is produced by lasers, LEDs, and certain types of lamps.
- **Polychromatic light**, on the other hand, consists of multiple wavelengths or colors. This type of light is produced by most light sources, including the sun, incandescent bulbs, and fluorescent lamps.
- Monochromatic light is often used in scientific and medical applications, such as spectroscopy and microscopy, where precise control of the wavelength of light is required.
- Polychromatic light is more commonly used in everyday applications, such as lighting and photography, where a broad spectrum of colors is desired.
- The human eye is more sensitive to polychromatic light, as it contains multiple wavelengths that stimulate the different types of photoreceptor cells in the retina.
- Monochromatic light can appear less bright than polychromatic light of the same intensity, as it only stimulates one type of photoreceptor cell.

CO-2 Compute and analyze various electrical and electronic properties of a given

1. **Resistance:** The opposition to the flow of electric current through a conductor. It is measured in ohms (Ω) and can be calculated using Ohm's Law, which states that the resistance is equal to the voltage divided by the current.
2. **Capacitance:** The ability of a system to store an electric charge. It is

measured in farads (F) and is determined by the distance between the plates of a capacitor, the area of the plates, and the dielectric constant of the material between the plates.

3. **Inductance:** The property of an electric conductor or circuit that causes an electromotive force to be generated by a change in the current flowing. It is measured in henries (H) and is determined by the number of turns in a coil, the cross-sectional area of the coil, and the magnetic permeability of the core material.
4. **Impedance:** The total opposition to the flow of alternating current in an electric circuit. It is measured in ohms (Ω) and is a complex quantity that takes into account both the resistance and the reactance of the circuit.
5. **Conductance:** The ease with which electric current flows through a conductor. It is the reciprocal of resistance and is measured in siemens (S).
6. **Reactance:** The opposition to the flow of alternating current in an electric circuit due to capacitance or inductance. It is measured in ohms (Ω) and is a complex quantity that varies with the frequency of the alternating current.
7. **Admittance:** The ease with which alternating current flows through an electric circuit. It is the reciprocal of impedance and is measured in siemens (S).
8. **Permittivity:** The measure of a material's ability to store an electric charge. It is determined by the dielectric constant of the material and is an important factor in the design of capacitors.
9. **Permeability:** The measure of a material's ability to conduct a magnetic field. It is determined by the magnetic properties of the material and is an important factor in the design of inductors and transformers.

Material Analysis by Using Various Experiments

Material analysis is the process of determining the chemical composition, structure, and properties of a material. This information is used to understand the behavior of the material under different conditions and to develop new materials with improved properties. There are several experiments that can be used to analyze materials, including:

1. **X-ray diffraction (XRD):** This technique is used to determine the crystal structure of a material. It involves shining X-rays onto a sample and measuring the diffraction pattern produced. The pattern provides information about the arrangement of atoms in the material.
2. **Scanning electron microscopy (SEM):** This technique is used to study the surface of a material. It involves scanning a focused beam of electrons across the surface of the sample and measuring the secondary electrons emitted. The resulting image provides information about the topography and composition of the surface.
3. **Transmission electron microscopy (TEM):** This technique is used to

study the internal structure of a material. It involves passing a beam of electrons through a thin sample and measuring the transmitted electrons. The resulting image provides information about the internal structure and composition of the material.

4. **Fourier transform infrared spectroscopy (FTIR):** This technique is used to determine the chemical composition of a material. It involves shining infrared light onto a sample and measuring the absorption of the light. The resulting spectrum provides information about the chemical bonds present in the material.

These are just a few examples of the many experiments that can be used to analyze materials. The choice of experiment depends on the specific information that is needed about the material.

CO-3 Verify different established laws with the help of optical and electrical

There are several established laws in the fields of optics and electricity that can be verified through experiments and observations. Some of these laws include:

1. **Ohm's Law:** This law states that the current flowing through a conductor is directly proportional to the voltage applied across it, provided the temperature remains constant. This can be verified by measuring the current and voltage across a conductor and plotting a graph to show the linear relationship.
2. **Snell's Law:** This law describes the relationship between the angles of incidence and refraction when light passes through a boundary between two different media. This can be verified by shining a light through a transparent medium and measuring the angles of incidence and refraction.
3. **Faraday's Law of Electromagnetic Induction:** This law states that a change in the magnetic field within a closed loop of wire will induce an electromotive force (EMF) in the wire. This can be verified by moving a magnet near a coil of wire and measuring the induced EMF.
4. **The Law of Reflection:** This law states that the angle of incidence is equal to the angle of reflection when light is reflected off a smooth surface. This can be verified by shining a light onto a smooth surface and measuring the angles of incidence and reflection.

These are just a few examples of the many established laws in the fields of optics and electricity that can be verified through experiments and observations. It is important to understand and verify these laws in order to have a deeper understanding of the principles that govern the behavior of light and electricity.

Experiments

An experiment is a scientific procedure undertaken to make a discovery, test a hypothesis, or demonstrate a known fact. Here are some key points to consider when conducting an experiment:

1. **Hypothesis:** A hypothesis is a proposed explanation for a phenomenon. It is important to have a clear and testable hypothesis before conducting an experiment.
2. **Variables:** Variables are factors that can be changed or controlled in an experiment. It is important to identify and control variables to ensure that the results of the experiment are accurate.
3. **Control group:** A control group is a group in an experiment that does not receive the treatment being tested. The control group is used as a baseline to compare the results of the treatment group.
4. **Data collection:** Data collection is the process of gathering and measuring information. It is important to have a clear and consistent method for collecting data to ensure that the results of the experiment are accurate.
5. **Data analysis:** Data analysis is the process of examining and interpreting the data collected during an experiment. It is important to use appropriate statistical methods to analyze the data and draw conclusions.
6. **Conclusion:** The conclusion is the final step in an experiment where the results are interpreted and the hypothesis is either accepted or rejected. It is important to clearly state the conclusion and provide evidence to support it.
7. **Replication:** Replication is the process of repeating an experiment to confirm the results. It is important to replicate experiments to ensure that the results are reliable and not due to chance.

Apply

- The term “apply” can have several meanings depending on the context in which it is used.
- In general, to apply means to make a formal request or to put something into operation or use.
- For example, one can apply for a job, a loan, or a permit by submitting an application or request to the appropriate authority.
- In the context of mathematics or science, to apply can mean to use a rule, principle, or formula to solve a problem or to understand a situation.
- In the context of cosmetics or medicine, to apply can mean to put a substance onto a surface, such as applying makeup to the face or applying ointment to a wound.
- In the context of law, to apply can mean to make a legal claim or to request a court to make a ruling or decision.

CO-4 Determine and calculate various physical properties of a given material by using

1. **Density:** The density of a material is defined as its mass per unit volume. It can be calculated by dividing the mass of the material by its volume.
2. **Elasticity:** Elasticity is the ability of a material to return to its original shape after being deformed. The modulus of elasticity, also known as Young's modulus, is a measure of the stiffness of a material and can be calculated by dividing the stress by the strain.
3. **Thermal conductivity:** Thermal conductivity is a measure of a material's ability to conduct heat. It can be calculated by dividing the heat flow rate by the temperature gradient.
4. **Electrical conductivity:** Electrical conductivity is a measure of a material's ability to conduct electricity. It can be calculated by dividing the current by the electric field.
5. **Hardness:** Hardness is a measure of a material's resistance to deformation. There are several methods to measure the hardness of a material, including the Brinell, Rockwell, and Vickers hardness tests.
6. **Tensile strength:** Tensile strength is the maximum stress that a material can withstand while being stretched or pulled before breaking. It can be calculated by dividing the maximum load by the cross-sectional area of the material.
7. **Compressive strength:** Compressive strength is the maximum stress that a material can withstand while being compressed before breaking. It can be calculated by dividing the maximum load by the cross-sectional area of the material.
8. **Shear strength:** Shear strength is the maximum stress that a material can withstand while being sheared before breaking. It can be calculated by dividing the maximum load by the cross-sectional area of the material.
9. **Fatigue strength:** Fatigue strength is the maximum stress that a material can withstand for a given number of cycles before breaking. It is typically determined through fatigue testing.
10. **Creep strength:** Creep strength is the maximum stress that a material can withstand for a given time period at a given temperature before undergoing significant deformation. It is typically determined through creep testing.

Various Experiments

There are numerous experiments that have been conducted in various fields of science. Some of the most notable experiments include:

1. **The Double-Slit Experiment:** This experiment, first performed by Thomas Young in 1801, demonstrated the wave-like behavior of light. It involved shining light through two parallel slits and observing the resulting interference pattern on a screen behind the slits.

2. **The Michelson-Morley Experiment:** This experiment, conducted by Albert Michelson and Edward Morley in 1887, was designed to detect the presence of a hypothetical medium called the “luminiferous ether” through which light was thought to propagate. The experiment failed to detect the ether, providing evidence for the theory of relativity.
3. **The Rutherford Gold Foil Experiment:** This experiment, conducted by Ernest Rutherford in 1911, involved firing alpha particles at a thin sheet of gold foil. The scattering of the alpha particles provided evidence for the existence of a small, dense nucleus at the center of the atom.
4. **The Stanford Prison Experiment:** This experiment, conducted by Philip Zimbardo in 1971, involved assigning college students to play the roles of prisoners and guards in a simulated prison environment. The experiment was intended to study the psychological effects of perceived power, but was terminated early due to the abusive behavior of the guards.
5. **The Milgram Experiment:** This experiment, conducted by Stanley Milgram in 1961, involved instructing participants to administer electric shocks to a “learner” (who was actually a confederate of the experimenter) in order to study obedience to authority. The experiment raised ethical concerns and is still widely discussed in psychology courses today.

These are just a few examples of the many experiments that have been conducted in the pursuit of scientific knowledge. Each experiment has contributed to our understanding of the world around us and has helped to shape the course of scientific inquiry.

Apply

1. The term “apply” can refer to the act of putting something into operation or use, such as applying a rule or principle.
2. It can also refer to the act of making a formal request or submission, such as applying for a job or a loan.
3. In the context of mathematics or science, “apply” can refer to the use of a formula, theorem, or principle to solve a problem or explain a phenomenon.
4. In general, to apply something means to put it into practical use or to make use of it in a specific situation.

CO-5 Study and estimate the performance and parameter of given equipment by using

1. To study and estimate the performance and parameter of given equipment, it is important to first understand the specifications and capabilities of the equipment.
2. This can be done by referring to the equipment’s manual or technical documentation, which should provide detailed information on its performance and parameters.

3. Additionally, conducting tests and experiments on the equipment can provide valuable data on its performance and capabilities.
4. This data can then be analyzed and used to estimate the equipment's performance and parameters.
5. It is also important to consider external factors that may affect the equipment's performance, such as environmental conditions and usage patterns.
6. By taking these factors into account, a more accurate estimation of the equipment's performance and parameters can be made.

Graphical and Computational Analysis

Graphical and computational analysis are two methods used to solve mathematical problems. Graphical analysis involves visualizing data or equations using graphs, while computational analysis involves using algorithms and computer programs to solve problems.

1. **Graphical Analysis:** Graphical analysis is a powerful tool for visualizing and understanding data. It involves creating graphs to represent data or equations, and then using these graphs to gain insights into the problem at hand. Graphical analysis can be used to identify trends, patterns, and relationships in data, and can also be used to solve equations and systems of equations.
2. **Computational Analysis:** Computational analysis involves using algorithms and computer programs to solve mathematical problems. This method is particularly useful for problems that are too complex to be solved by hand, or for problems that require a large amount of data to be processed. Computational analysis can be used to solve a wide range of problems, including optimization problems, statistical analysis, and numerical simulations.

Both graphical and computational analysis are important tools for solving mathematical problems, and are often used together to gain a deeper understanding of the problem at hand. While graphical analysis provides a visual representation of the problem, computational analysis provides a more rigorous and precise solution. Together, these two methods can be used to solve complex problems and gain insights into the underlying mathematics.

Apply

1. The term “apply” can refer to the act of putting something into operation or use.
2. It can also mean to make a formal request, such as applying for a job or a loan.
3. In the context of mathematics, “apply” can mean to use a mathematical operation or formula to a given set of values.
4. In the context of science, “apply” can mean to use a scientific principle or theory to explain or predict a phenomenon.

5. In general, “apply” can mean to put into practice or use something that has been learned or acquired.

Reference Books

Reference books are a type of non-fiction book that provides information on a specific subject. They are often used for research and study purposes. Some common types of reference books include:

1. **Encyclopedias:** These books provide comprehensive information on a wide range of subjects. They are organized alphabetically and contain articles on various topics.
2. **Dictionaries:** These books provide definitions and meanings of words. They are also organized alphabetically and can be used to look up the spelling and pronunciation of words.
3. **Atlases:** These books contain maps and geographical information. They can be used to find the location of a place or to learn about the geography of a region.
4. **Almanacs:** These books contain information on a wide range of subjects, including current events, statistics, and historical facts. They are published annually and provide a snapshot of the world at a specific point in time.
5. **Handbooks and Manuals:** These books provide practical information and instructions on a specific subject. They can be used to learn how to do something or to find out how something works.

Reference books are an important resource for students and researchers. They provide reliable and accurate information that can be used to learn about a subject or to support an argument. They are available in print and digital formats and can be found in libraries and bookstores.

1. Practical Physics- K. K. Dey & B. N. Dutta (Kalyani Publishers New Delhi)

- Practical Physics is a book written by K. K. Dey and B. N. Dutta.
- It is published by Kalyani Publishers, New Delhi.
- The book is a comprehensive guide to the practical aspects of physics.
- It covers a wide range of topics, including mechanics, heat, light, sound, electricity, and magnetism.
- The book is written in a clear and concise manner, making it easy to understand for students.
- It includes numerous examples, illustrations, and exercises to help students apply the concepts they have learned.
- Practical Physics is an essential resource for students studying physics at the undergraduate level.

2. Engineering Physics-Theory and Practical- Katiyar & Pandey (Wiley India)

- “Engineering Physics-Theory and Practical” is a book written by Katiyar & Pandey and published by Wiley India.
- The book covers the fundamental concepts of physics and their applications in engineering.
- It is designed to provide a comprehensive understanding of the subject to engineering students.
- The book is divided into two parts: theory and practical.
- The theory part covers topics such as mechanics, waves and oscillations, optics, electromagnetism, and modern physics.
- The practical part includes experiments and activities that help students apply the concepts learned in the theory part.
- The book is written in a clear and concise manner, making it easy for students to understand the subject.
- It is a useful resource for students preparing for exams or for anyone interested in learning more about the subject of engineering physics.

3. Engineering Physics Practical- S K Gupta (Krishna Prakashan Meerut)

- “Engineering Physics Practical” is a book written by S K Gupta and published by Krishna Prakashan Meerut.
- The book is intended for students studying engineering and covers practical aspects of physics.
- It includes experiments and practical exercises to help students understand the concepts of physics in an applied context.
- The book is written in a clear and concise manner, making it easy for students to follow and understand.
- It is a valuable resource for students preparing for exams or for anyone looking to improve their understanding of practical physics.